

# Embodied Language Models

From multi-task learning towards the generalist robot

Thomas Doms, Alexander Belousov

# Context

Research project 2 (12 sp)

- Transformers United
- **Deep multi-task & meta-learning**

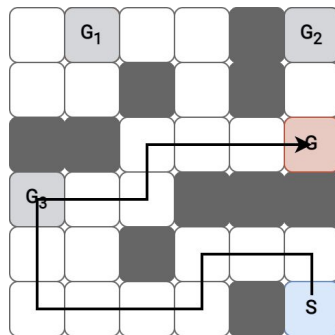
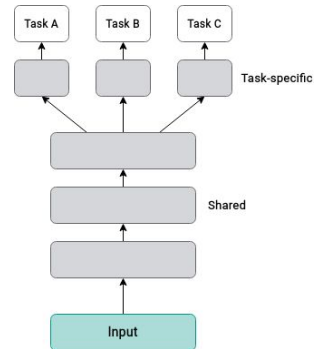
## CS 330: Deep Multi-Task and Meta Learning



# Content

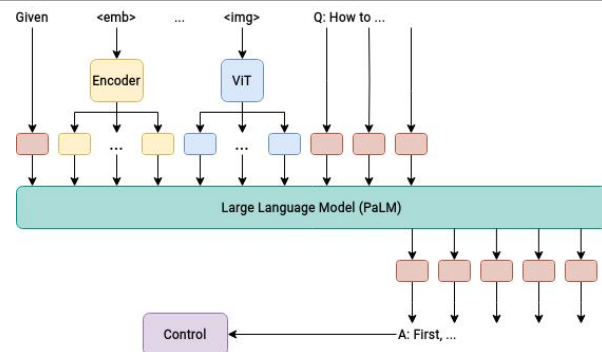
A brief overview of this lecture

## The Basics



## Reinforcement Learning

## Language Conditioning



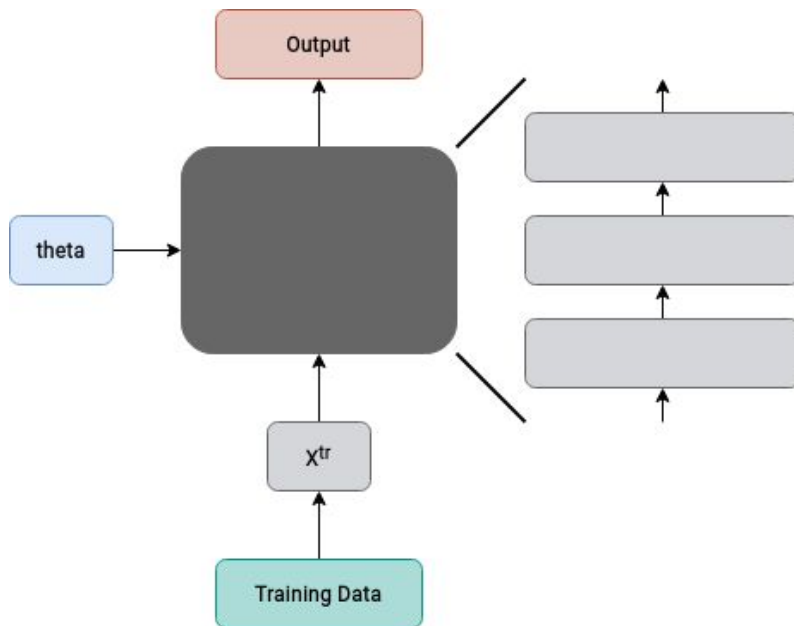
# Multi-Task Learning

# Definition

Let's refresh the basics

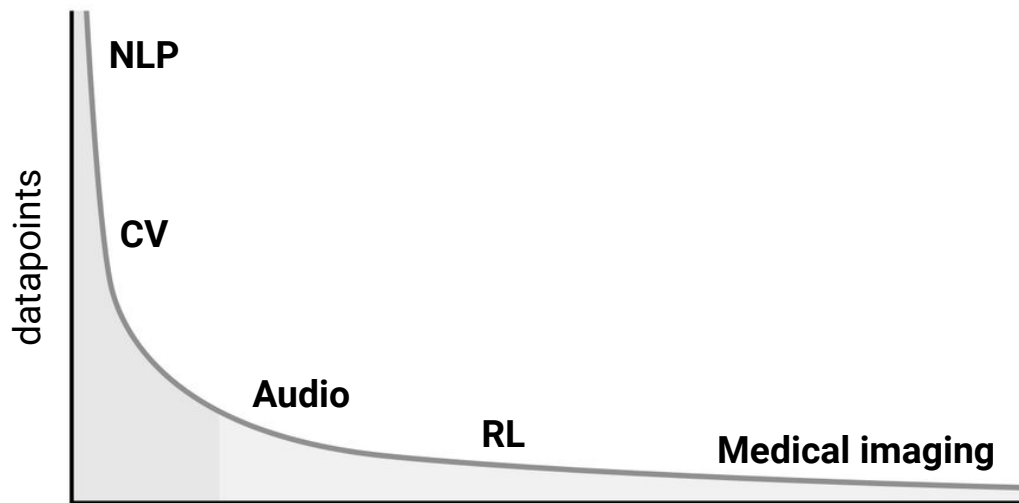
**Supervised learning**

$$\min_{\theta} \mathcal{L}(\theta, \mathcal{D}) \text{ where } \mathcal{D} = \{(x, y)_k\}$$



# Motivation

Some data modalities are very hard to acquire

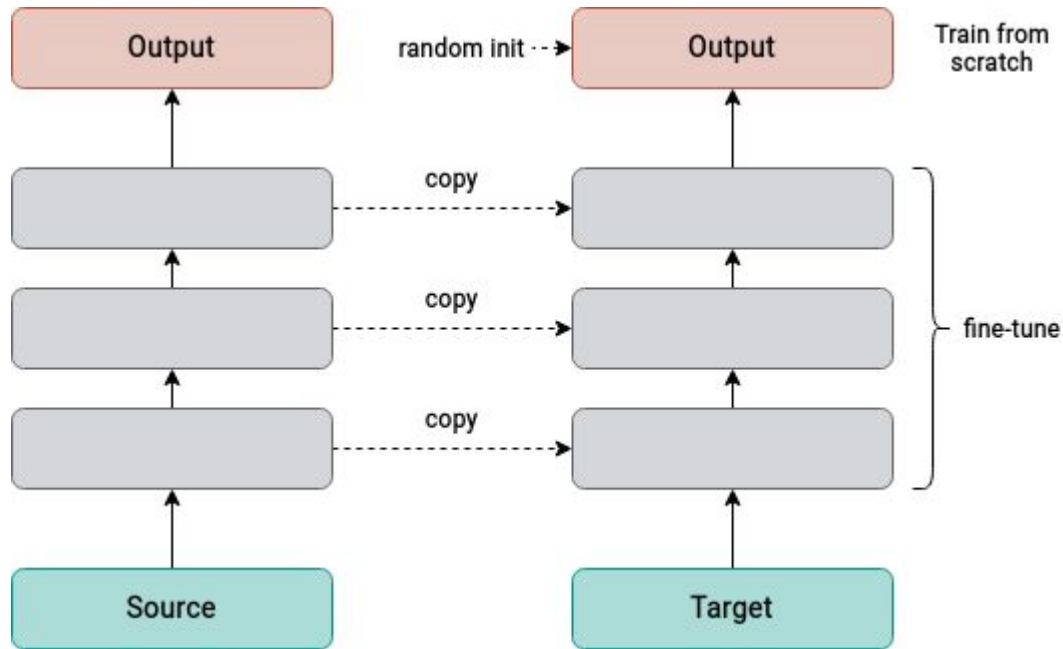


# Transfer learning

Solve target task  $\mathcal{T}_b$

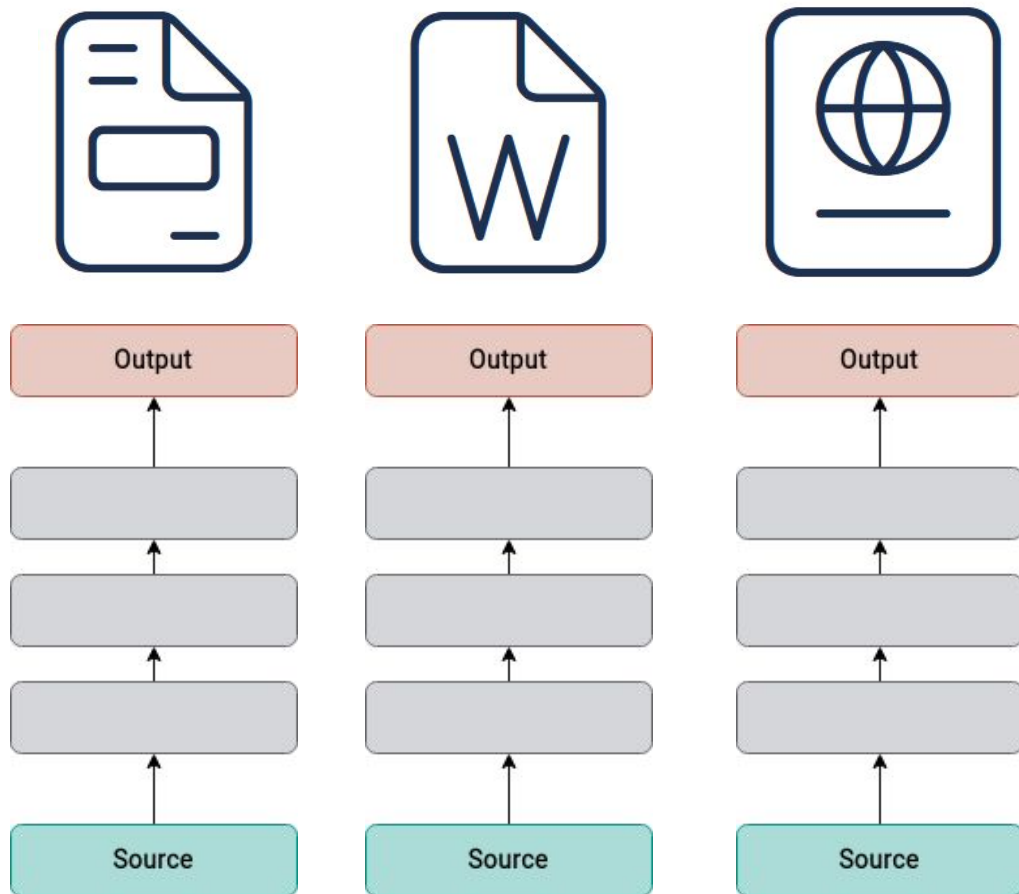
By transferring knowledge learned from  $\mathcal{T}_a$

Without access to  $\mathcal{D}_a$



# Limitations

Fine-Tuning can Distort Pretrained  
Features and Underperform  
Out-of-Distribution  
*Kumar Ananya et al. (2022)*



# Multi-task learning

CS330: Deep multi-task & meta-learning  
*Chelsea Finn (2021)*

## Definitions

Task:  $\mathcal{T}_i \triangleq (\mathcal{D}_i, \mathcal{L}_i)$

Objective ( $L_1$ ):  $\min_{\theta} \sum_{i=1}^T \mathcal{L}_i(\theta, \mathcal{D}_i)$

## Example

$$\alpha \sum_{i \in I} \mathcal{L}_i(\theta_i, \mathcal{D}_i) + (1 - \alpha) \sum_{e \in E} \mathcal{L}_e(\theta_e, \mathcal{D}_e)$$

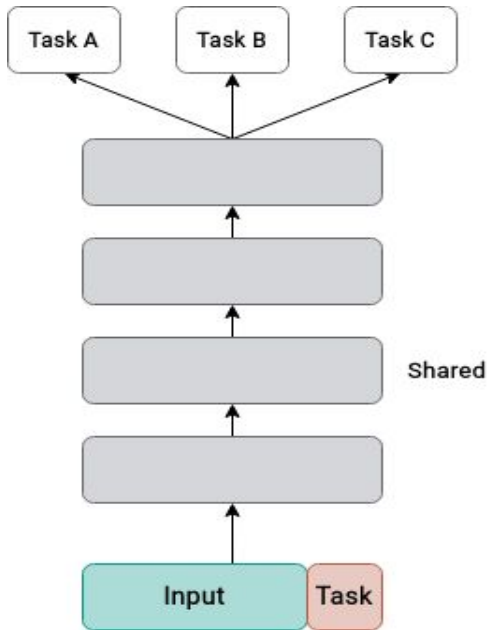
## Assumption

Tasks share some structure  $\rightarrow$  most often, they do

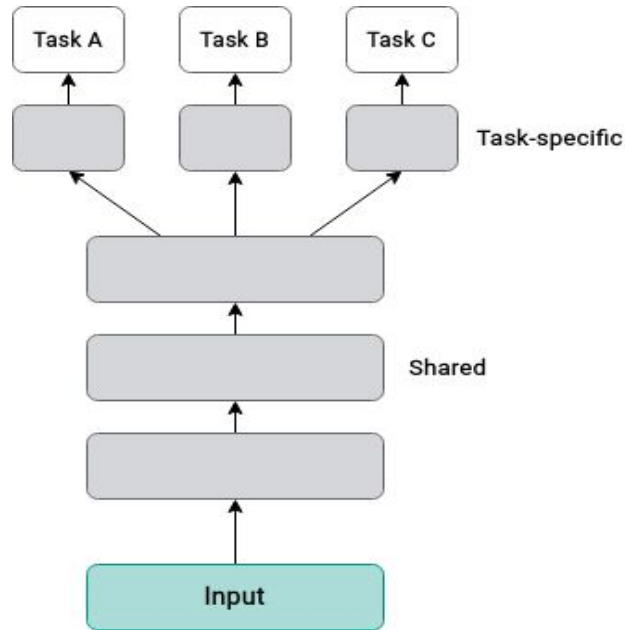
Learning this structure is beneficial for both tasks

# Multi-task architectures

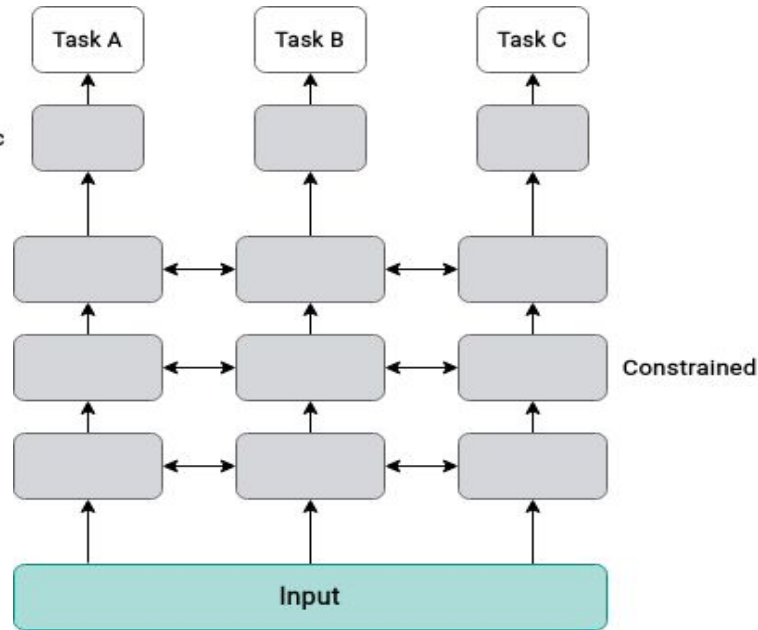
**All Sharing**



**Hard Sharing**



**Soft Sharing**



# Summary

Short summary of multi-task learning

## **Train multiple tasks together**

No tradeoff between specificity and generality

Less overfitting

Higher accuracy

# Meta learning

# Learning to Learn

Who can figure it out?

$$4 \text{ 🍕 } 6 = ?$$

$$3 \text{ 🍕 } 5 = 18$$

$$1 \text{ 🍕 } 2 = 3$$

$$2 \text{ 🍕 } 3 = 8$$

$$6 \text{ 🍕 } 1 = 12$$

# Definitions

Learning to Learn with Gradients  
*Finn, Chelsea (2018)*

## Mathematically

Given data from  $\mathcal{T}_1, \dots, \mathcal{T}_n$ , quickly solve new task  $\mathcal{T}_{\text{test}}$

Supervised learning:  $\min_{\theta} \mathcal{L} (f_{\theta}(X^{tr}), Y^{tr})$

Meta-learning:  $\min_{\theta} \sum_i \mathcal{L}_i (f_{\theta} (\mathcal{D}_i^{tr}, \mathcal{D}_i^{test}))$

## Intuitively

Find the set of parameters  $\theta$  such that new tasks can be learned quickly.

# Support & Query

Support



1

2

1

2



1

2

1

2



1

2

1

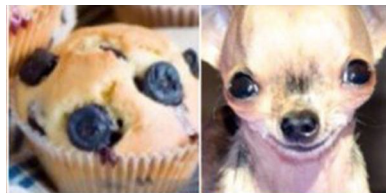
2

Query



1

2



1

2



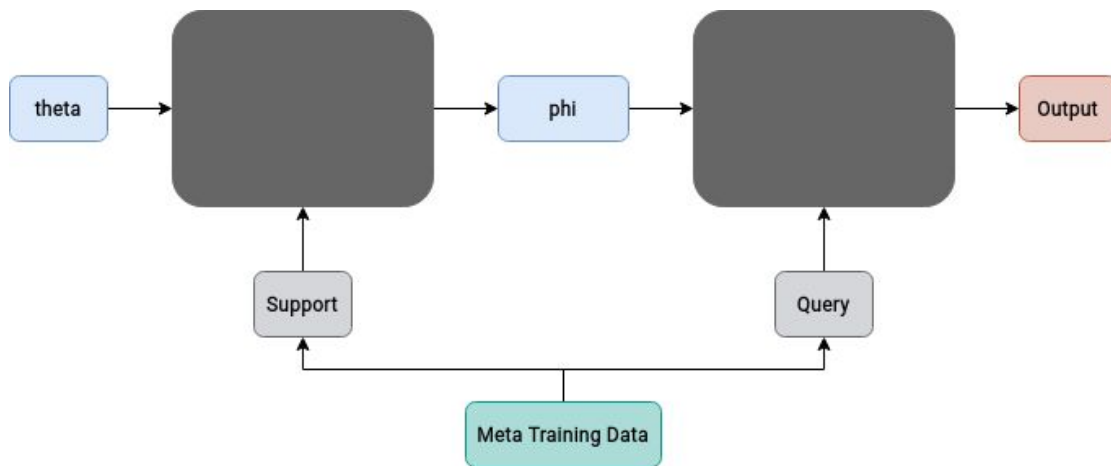
?

?

# Black-box meta learning

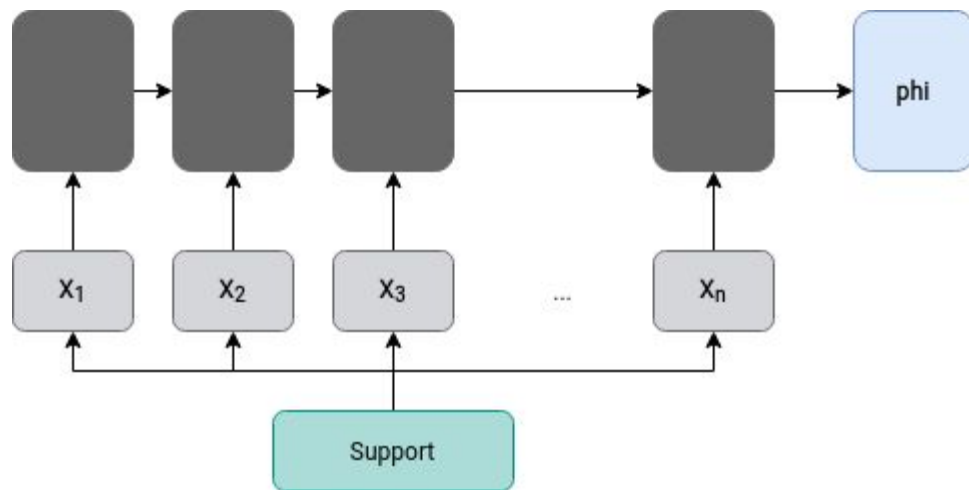
$$\phi = f_{\theta}(D^{tr})$$

$$\hat{y} = g_{\phi}(D^{test})$$



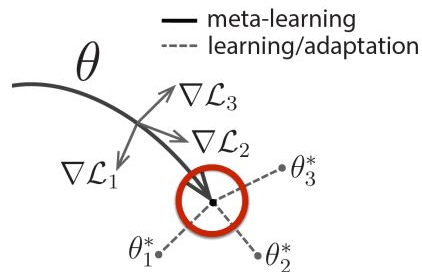
# Black-box meta learning

A concrete example using RNNs



# Optimisation based meta learning

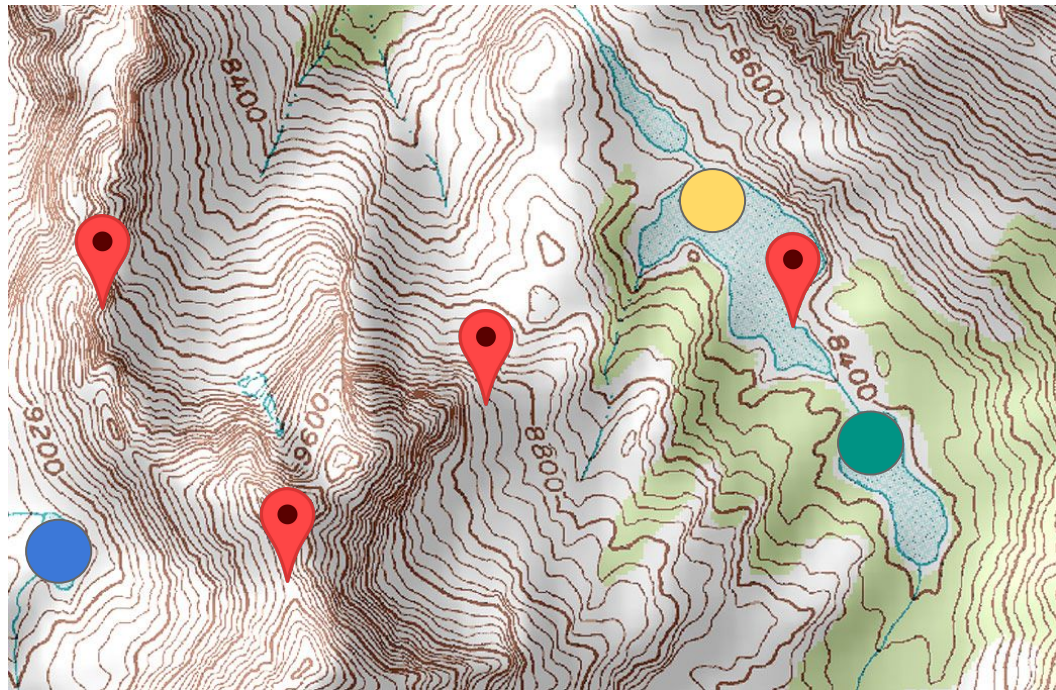
Model-Agnostic Meta-Learning for Fast  
Adaptation of Deep Networks  
*Finn, Chelsea et al. (2017)*



$$\phi = f(\theta, D^{tr}) = \theta - lr \cdot \nabla_{\theta} L(\theta, D^{tr})$$

# Quiz Time

Which marker is the best candidate for  $\theta$ ?



# Summary

What kind of learning algorithms have we discussed?

## Transfer learning

Solve  $\mathcal{T}_b$  by transferring knowledge from  $\mathcal{T}_a$

## Multi-task learning

Solve multiple tasks  $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_n$  at once

## Meta-learning

Given data from  $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_n$  quickly solve  $\mathcal{T}_{\text{test}}$

# Questions?

# Reinforcement Learning

# Multi-Task RL

Why multi-task?

## Cross-task generalization

Performance increases for all tasks



# Multi-Task RL

Why multi-task?

**Cross-task generalization**

**Easier exploration**

Tasks share knowledge



# Multi-Task RL

Why multi-task?

**Cross-task generalization**

**Easier exploration**

**Sequencing for long-horizon tasks**

Long tasks can be split into easier sub-tasks



# Multi-Task RL

Why multi-task?

**Cross-task generalization**

**Easier exploration**

**Sequencing for long-horizon tasks**

**Reset-free learning**

No intervention needed

Generalize to different starting states and goals

# Multi-Task RL

Why multi-task?

**Cross-task generalization**

**Easier exploration**

**Easier exploration**

**Sequencing for long-horizon tasks**

**Reset-free learning**

**Per-task sample-efficiency gains**

Fewer examples per task needed

# Multi-Task RL

## Task Specification

**Task can be defined by**

New State/Action space

Different dynamics

Different reward function

$$\mathcal{T}_i = \{S_i, A_i, p_i(s_1), p_i(s'|s, a), r_i(s, a)\}$$

**(Optional) Task identifier inside state**

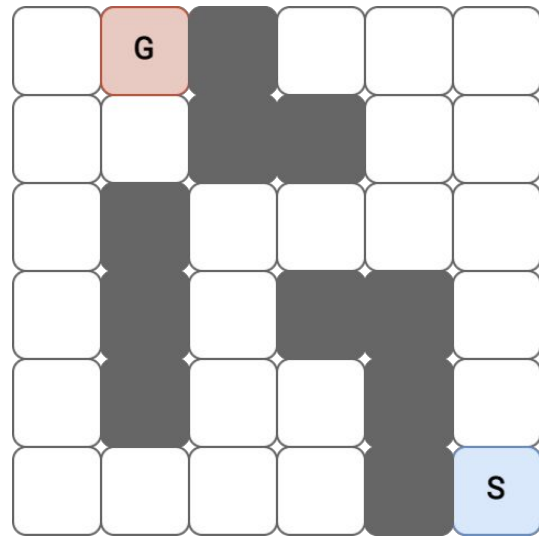
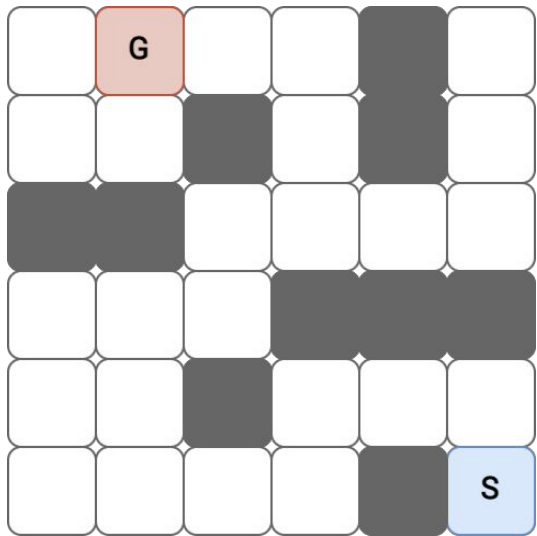
One-hot

Language description

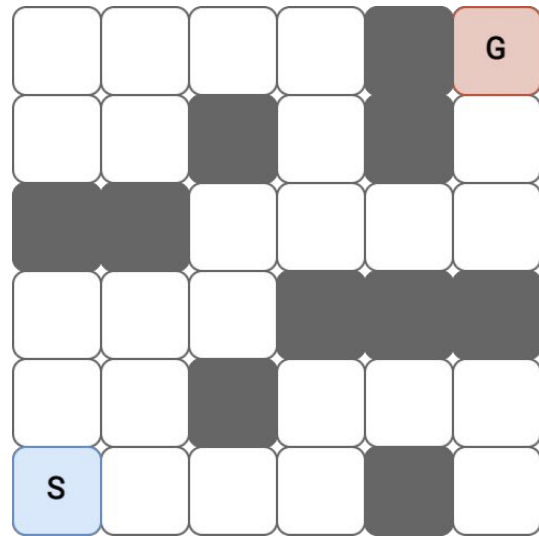
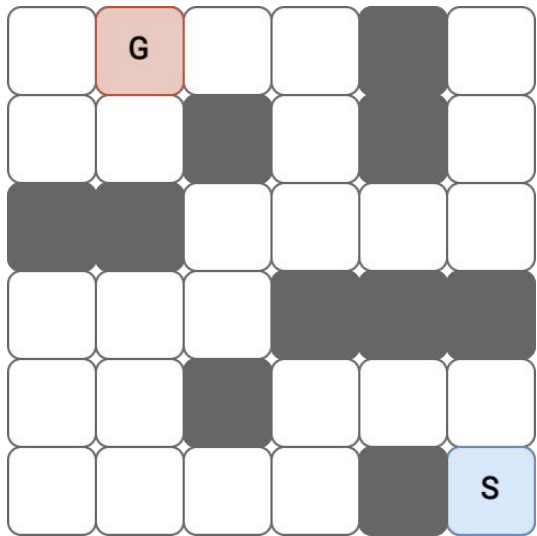
Goal state  $\Rightarrow$  Goal-conditioned RL

$$s = (\bar{s}, z_i)$$

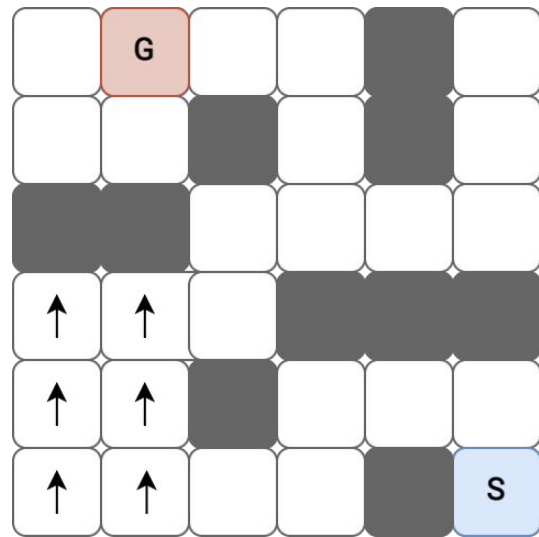
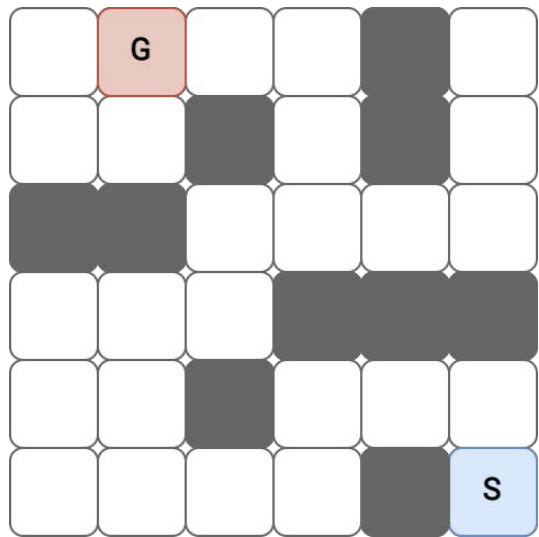
# Defining tasks



# Defining tasks



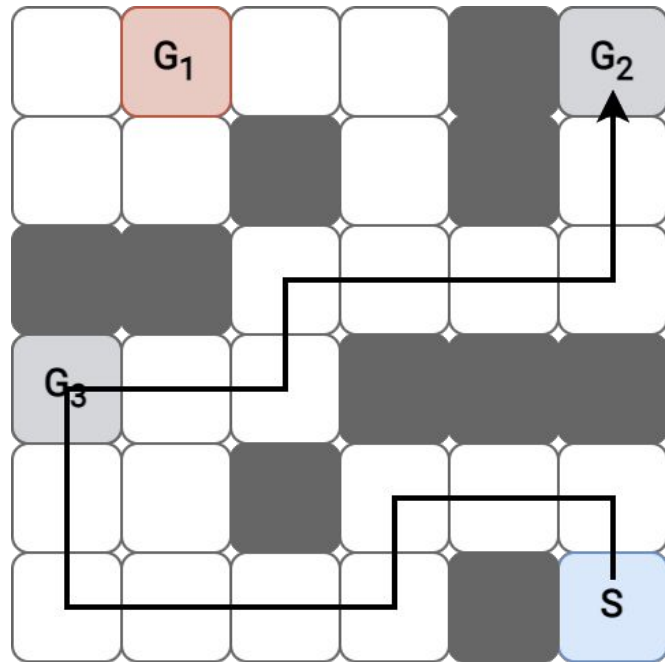
# Defining tasks



# Multi-Task RL

Hindsight Relabeling  
or Hindsight Experience Replay (HER)

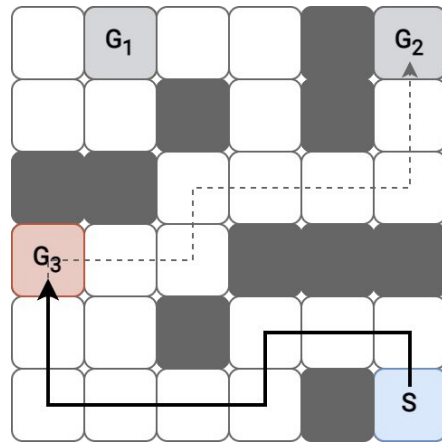
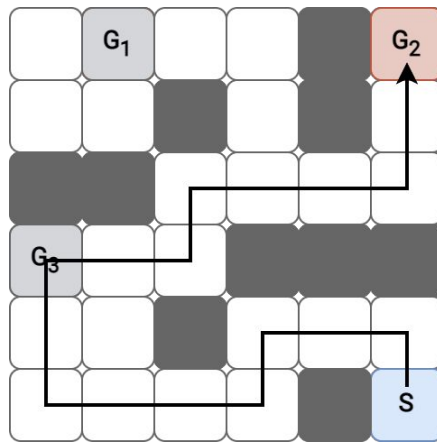
Hindsight experience replay  
*Andrychowicz, Marcin, et al. (2017)*



# Multi-Task RL

Hindsight Relabeling  
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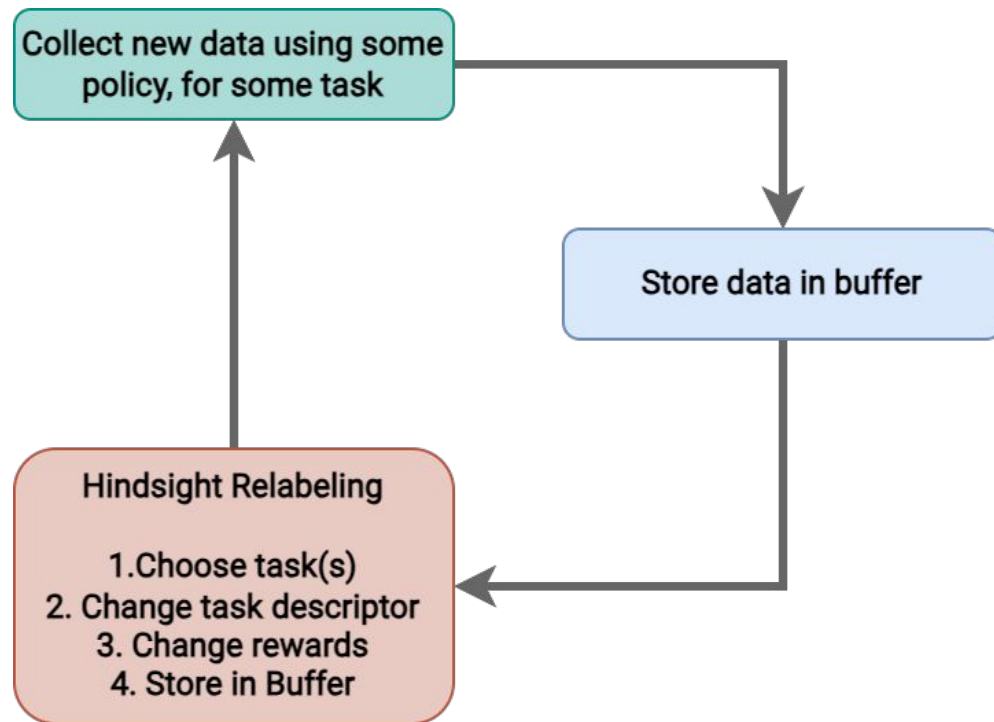
Hindsight experience replay  
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# Multi-Task RL

Hindsight Relabeling  
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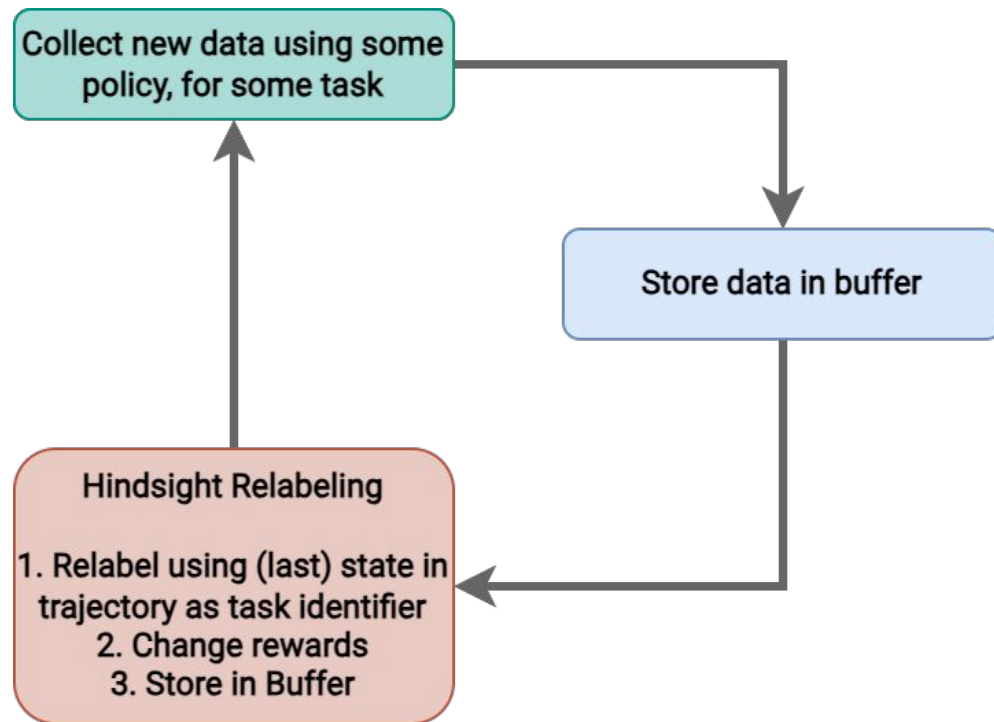
Hindsight experience replay  
*Andrychowicz, Marcin, et al. (2017)*



# Goal-Conditioned RL

Pretend achieved goal is what we  
wanted to do, even if real goal not  
reached

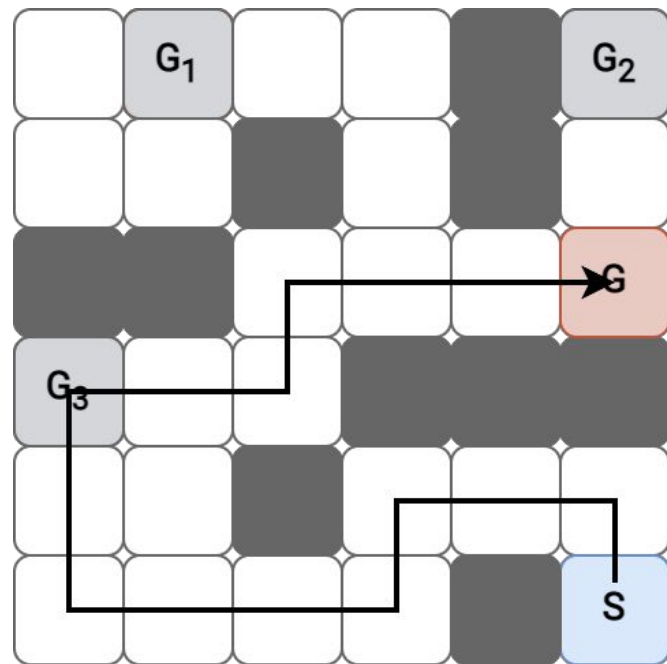
Hindsight experience replay  
*Andrychowicz, Marcin, et al. (2017)*



# Goal-Conditioned RL

Pretend achieved goal is what we  
wanted to do, even if real goal not  
reached

Hindsight experience replay  
*Andrychowicz, Marcin, et al. (2017)*



# Goal-Conditioned RL

Actionable models: Unsupervised offline  
reinforcement learning of robotic skills

*Chebatar, Yevgen, et al. (2021)*

Goal image



Robot execution



# Goal-Conditioned RL

Pretend achieved goal is what we  
wanted to do, even if real goal not  
reached

Use **any state as goal** of trajectory, not just final

Many more samples = more exploration

## **Always optimal data**

Goal is always reached after relabeling

Use unstructured dataset

→ **Human play**

→ Turn into supervised dataset

→ **Imitation Learning**

# Meta-RL

## Problem Statement

### Inputs

Current state  $s_t$

K previous timesteps or rollouts from policy

### Outputs

Action  $a_t$

$$a_t = f_{\theta}(\mathcal{D}^{tr}, s_t)$$

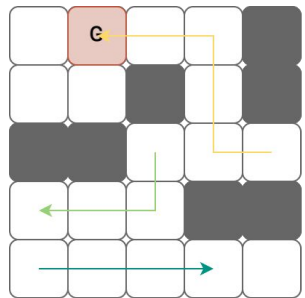
### Goal

Given small amount of experience

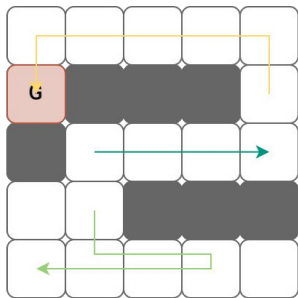
Adapt to new task

# Meta RL example

Meta Train

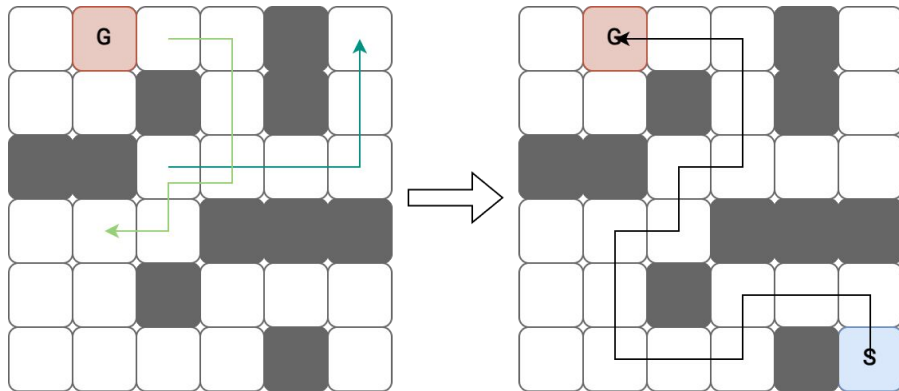


$\mathcal{T}_1$



$\mathcal{T}_2$

Meta Test



$\mathcal{D}_j^{train}$

# Meta-RL

## Black-Box Meta-RL

### Training

1. Sample Task  $T_i$
2. Roll-out policy  $\pi(a|s, \mathcal{D}_i^{tr})$  for N episodes
3. Store sequence in buffer  $\mathcal{D}^{tr}$  for task  $T_i$
4. Update policy to maximize discounted return for all tasks
5. Repeat

- + General & expressive
- + Variety in design choices/architectures
- Hard to optimize
- ~ Inherits sample efficiency from RL optimizer

# Meta-RL

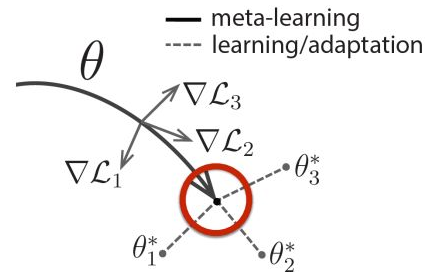
## Optimisation-Based Meta-RL

### Training

1. Sample task  $T_i$
2. Collect  $\mathcal{D}_i^{tr}$  by rolling out  $\pi_\theta$
3. Inner loop adaptation
$$\phi_i = \theta + \alpha \nabla_\theta J_i(\theta)$$
4. Collect  $\mathcal{D}_i^{ts}$  by rolling out  $\pi_{\phi_i}$
5. Outer loop update

$$\theta \leftarrow \theta + \beta \sum_{task\ i} \nabla_\theta J_i(\phi_i)$$

6. Repeat



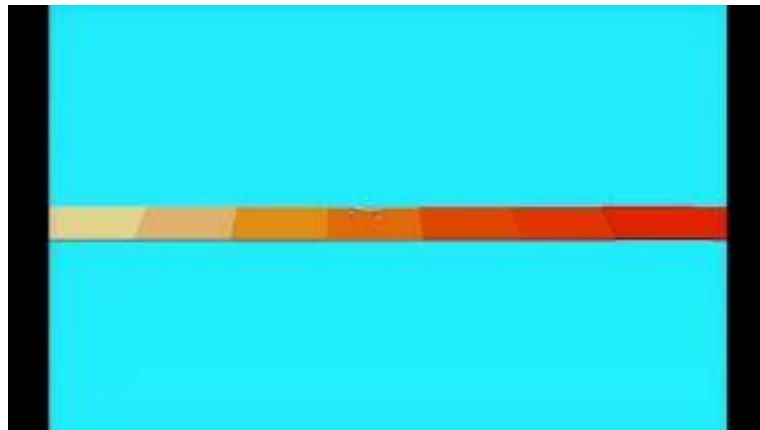
# Meta-RL

## Optimization-Based Meta-RL

Learning to Adapt in Dynamic  
Environments through Meta-RL.

*Nagabandi\*, Clavera\*, Liu, Fearing,  
Abbeel, Levine, Finn. (2019)*

- + Inductive bias
- + Easy to combine with policy gradients, model-based methods
- Hard to combine with value-based methods
- Policy gradients are very noisy
- ~ Inherits sample efficiency from outer RL optimizer



Questions?

# Language Models

# Language Prediction

Improving Language Understanding by  
Generative Pre-Training  
*Radford, Alec et al. (2018)*

**The Eiffel tower is in Paris**

| $x_i$                  | $y_i$  |
|------------------------|--------|
| The Eiffel tower is in | Paris  |
| The Eiffel tower is    | in     |
| The Eiffel tower       | is     |
| The Eiffel             | tower  |
| The                    | Eiffel |
|                        | The    |

# Transformers

Improving Language Understanding by  
Generative Pre-Training  
*Radford, Alec et al. (2018)*

## Sequential data

A stream of tokens

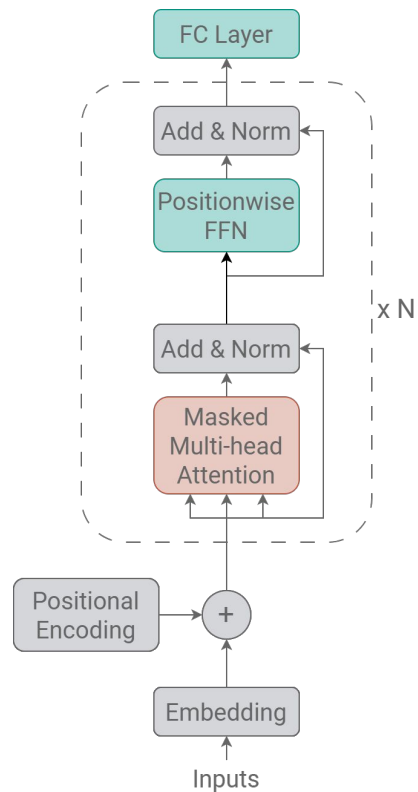
Can vary in length

## Three main components

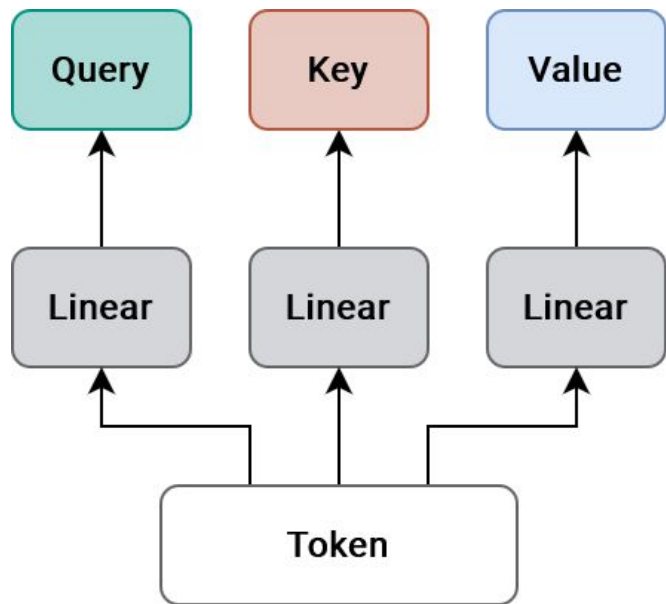
Positional encoding

Attention mechanism

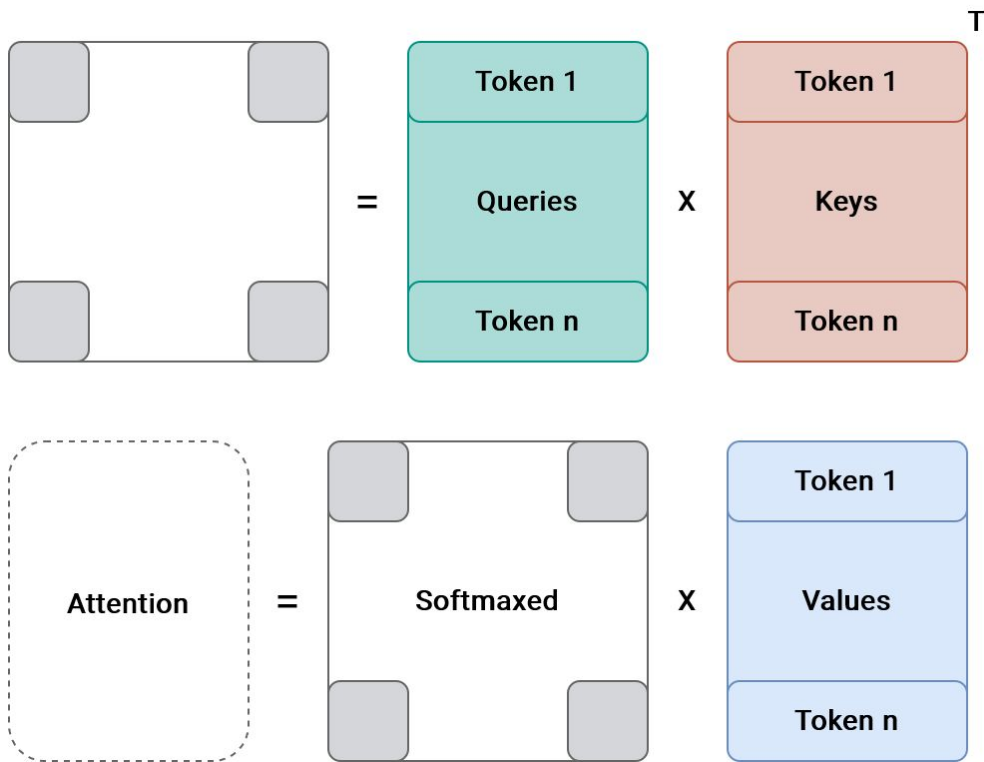
Positionwise FFN



# Attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



# Encoders & Decoders

Attention is all you need  
*Vaswani, Ashish et al. (2017)*

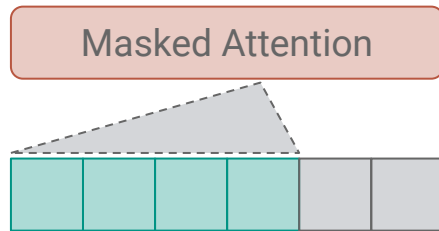
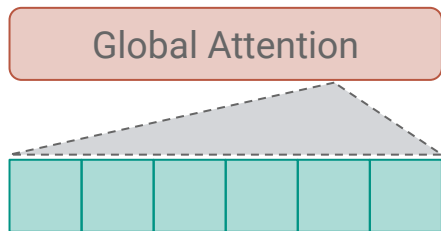
**Decoders** → look at previous tokens

Useful for text generation

Does not leak future tokens

**Encoders** → look at all tokens

Useful for text classification (sentiment analysis)



# Cross-Attention

Attention is all you need  
*Vaswani, Ashish et al. (2017)*

## Self-attention

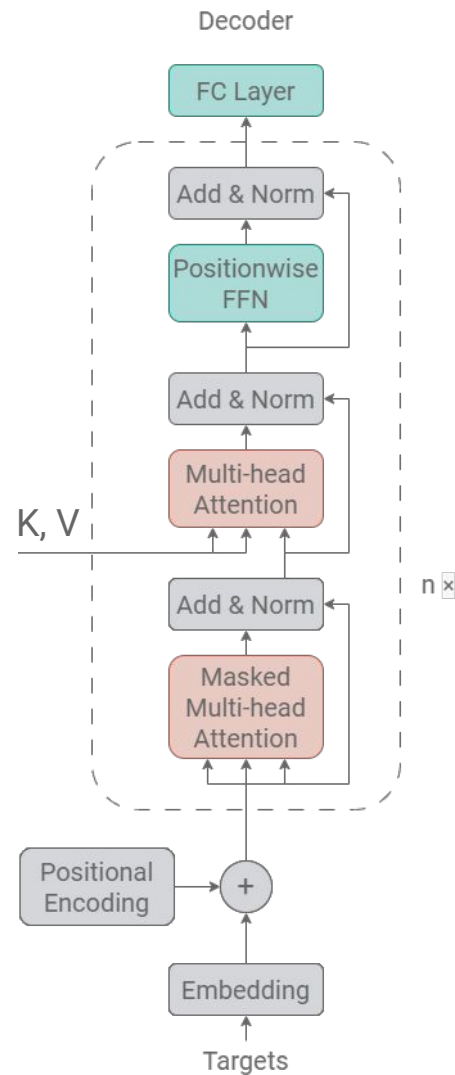
Query the data itself

Keys & values from original tokens

## Cross-attention

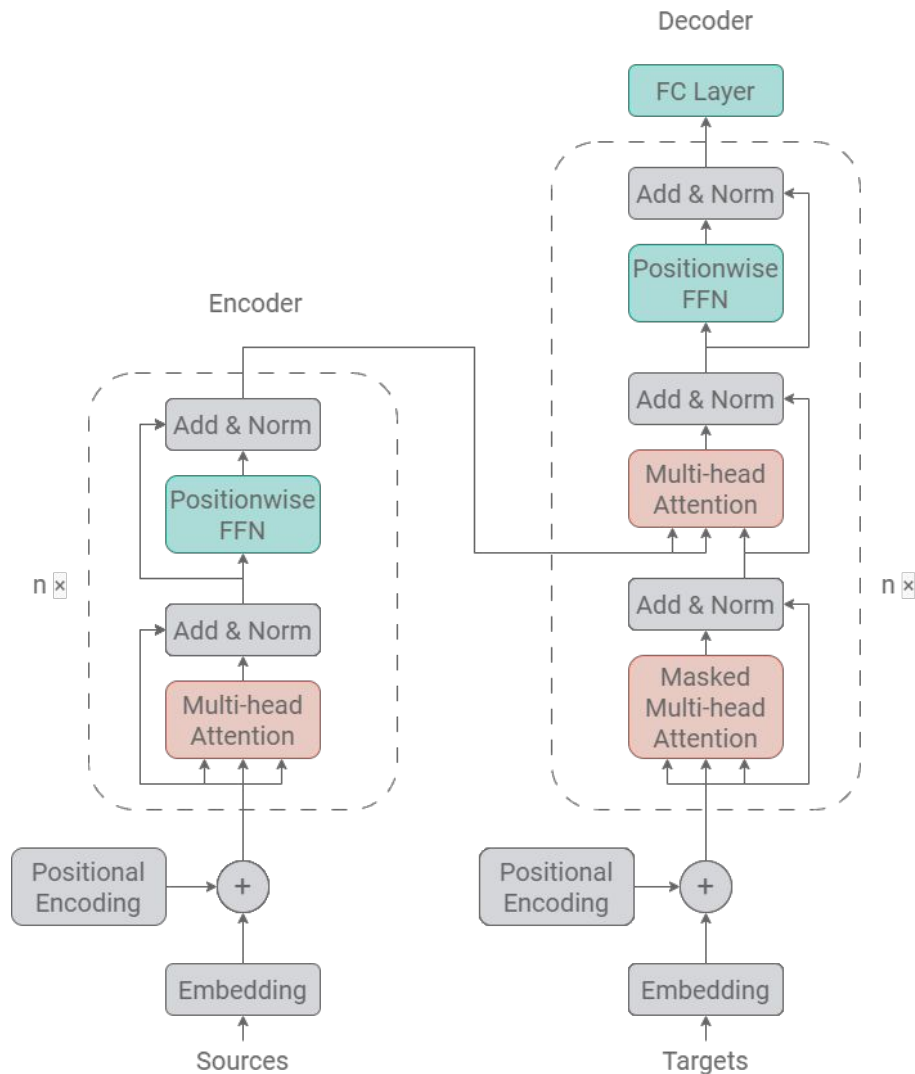
Query other keys and values

Keys & values source can vary



# The Full Transformer

Attention is all you need  
*Vaswani, Ashish et al. (2017)*



# Foundation models

- Trained on huge amounts of data
- Largest models in existence
- Are very versatile

PaLM: Scaling Language Modeling with Pathways

*Chowdhery, Aakanksha et al. (2022)*

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Total dataset size = 780 billion tokens

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| Data source                               | Proportion of data |
|-------------------------------------------|--------------------|
| Social media conversations (multilingual) | 50%                |
| Filtered webpages (multilingual)          | 27%                |
| Books (English)                           | 13%                |
| GitHub (code)                             | 5%                 |
| Wikipedia (multilingual)                  | 4%                 |
| News (English)                            | 1%                 |

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# Foundation Model Overview

| Name       | Company  | Release Date | Number of Parameters | Corpus Size  |
|------------|----------|--------------|----------------------|--------------|
| GPT-2      | OpenAI   | 2019         | 1.5 billion          | 10 billion   |
| GPT-3      | OpenAI   | 2020         | 175 billion          | 300 billion  |
| PaLM       | Google   | 2022         | 540 billion          | 768 billion  |
| GPT-4      | OpenAI   | 2023         | ~ 1 trillion*        | unknown      |
| Chinchilla | DeepMind | 2022         | 70 billion           | 1.4 trillion |
| Llama      | Meta     | 2023         | 65 billion           | 1.4 trillion |
| PaLM 2     | Google   | 2023         | 340 billion          | 3.5 trillion |

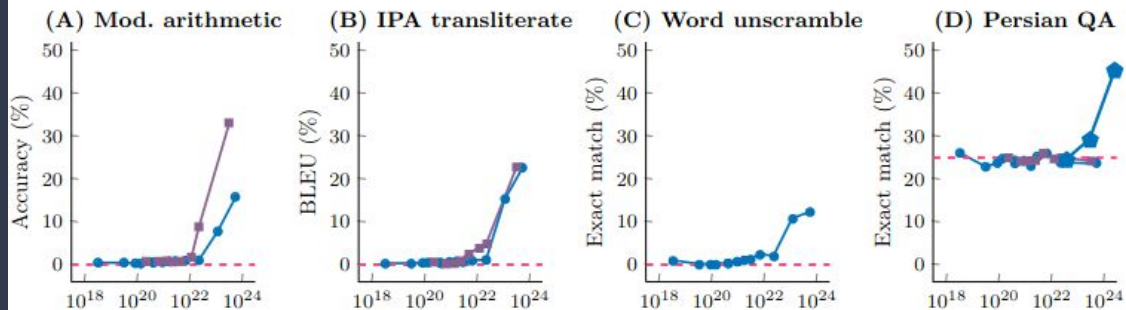
\*not actually known

# Emergent Abilities

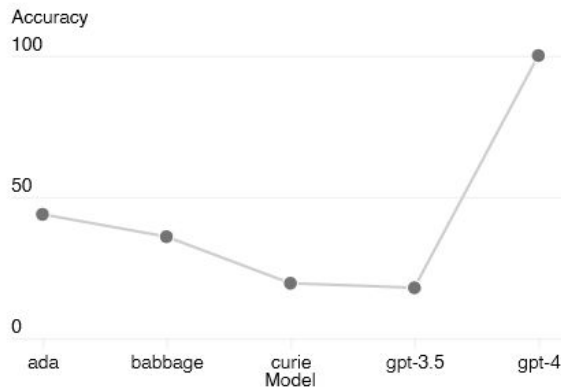
Emergent Abilities of Large Language Models

*Wei, Jason et al. (2022)*

GPT-4 Technical Report  
*OpenAI (2023)*



## Inverse scaling prize, hindsight neglect



# Few-Shot Abilities

Language Models are Few-Shot Learners

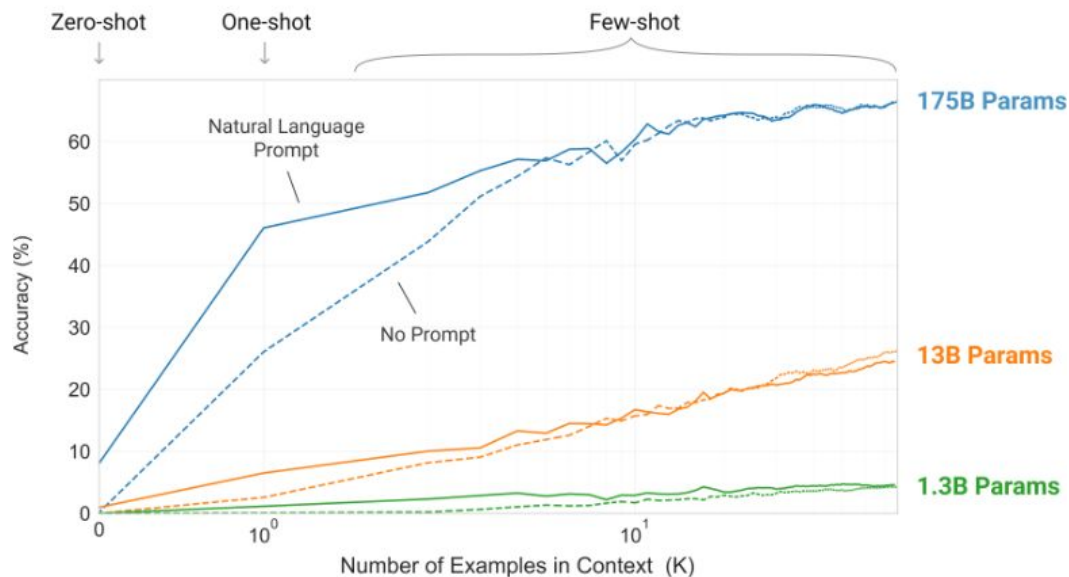
*Brown, Tom et al. (2020)*

Fine-tuned Language Models are Zero-Shot Learners

*Wei, Jason et al. (2022)*

## In-Context Learning (ICL)

“Wish You Were Here: 1979, The Dark side of the Moon:”



# Attention as a Meta-Optimiser

A Survey on In-context Learning  
*Dong, Qingxiu et al. (2023)*

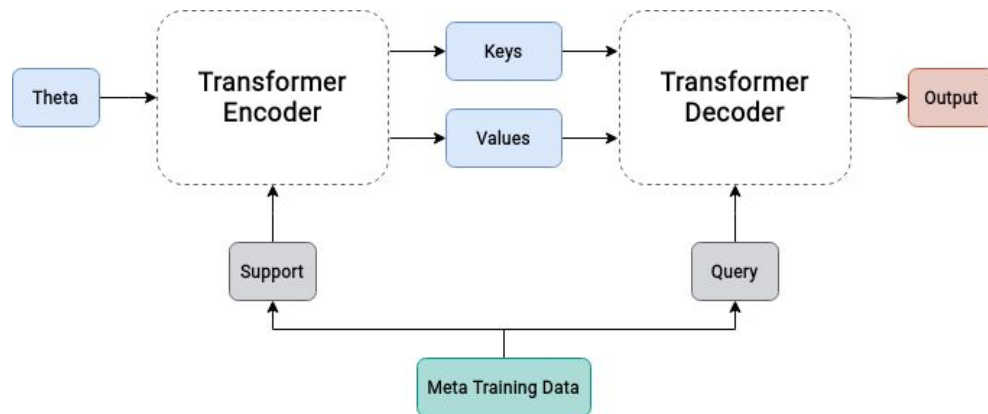
Language Models Implicitly Perform  
Gradient Descent as  
Meta-Optimizers  
*Dai, Damai et al. (2023)*

**Attention changes the latent space like fine-tuning (FT)**

$$(W_0 + \Delta W)x \approx W_0x + \text{Attention}(C)$$

**These modifications resemble fine-tuning**

Latent variables show high similarity between ICL and FT



# Fine-Tuning

Training language models to follow instructions with human feedback

*Ouyang, Long et al. (2022)*

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

*Devlin, Jacob et al. (2018)*

## Transfer knowledge to specific domains

Ask open-ended question on a set of documents

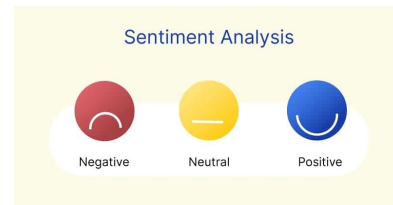
## Models retain insight of language

Knowledge of idioms

## Models retain the few-shot capabilities and ICL

Respond correctly to never-seen inputs

Tolerance to slight distribution shift



# Language Conditioning

# Goal

Interactive Language: Talking to Robots  
in Real Time

*Lynch, Corey et al. (2022)*

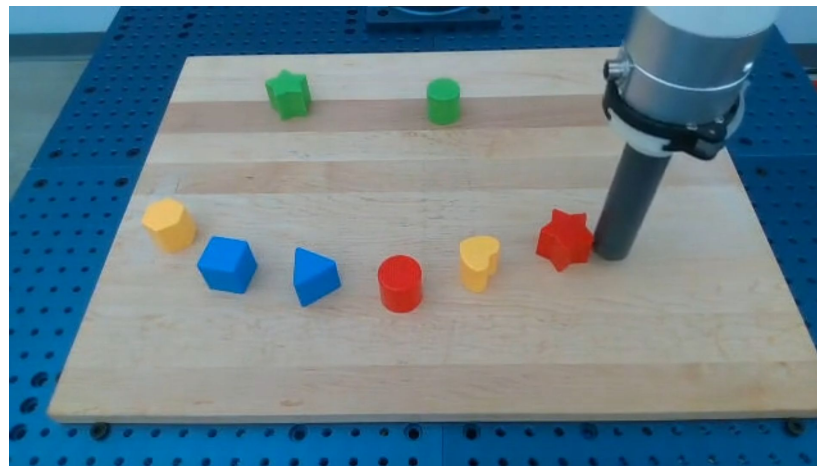
## **Guide robot using natural language**

Complete long-term objectives

Give short-term instructions

Rectify mistakes

## **Free vocabulary**



# Dataset Gathering

Interactive Language: Talking to Robots  
in Real Time

*Lynch, Corey et al. (2022)*

## Play dataset

Human operators

Optional objective prompts

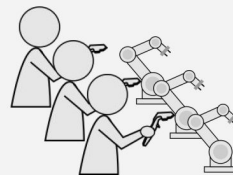
Explore the environment as much as possible

Actions are better than random

Prompts discarded  $\Rightarrow$  Unlabeled dataset



teleoperated collect



2.7k  
hours

# Dataset Gathering

Interactive Language: Talking to Robots  
in Real Time

*Lynch, Corey et al. (2022)*

## Event-selectable Hindsight Relabeling

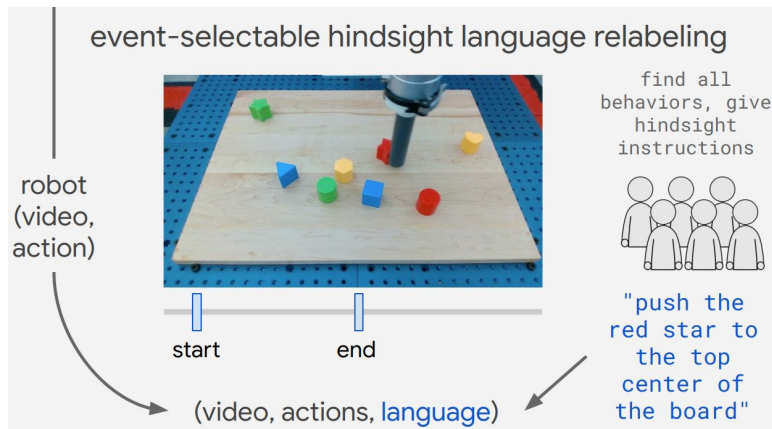
Select few second fragment in replay

→ Better than using random windows

Add natural language annotation

→ Open vocabulary

Annotations ⇒ Labeled dataset



# Imitation Learning

Interactive Language: Talking to Robots  
in Real Time

*Lynch, Corey et al. (2022)*

**Always-optimal labeled dataset**

Use simple objectives

**Base imitation learning objective**

$$\min_{\theta} \sum_{s, a \sim \mathcal{D}} ||a, \pi_{\theta}(s)||_2^2$$

**Condition on language**

$$\min_{\theta} \sum_{s, a, l \sim \mathcal{D}} ||a, \pi_{\theta}(s, l)||_2^2$$

# Architecture: LAVA (Language Attends to Vision to Act)

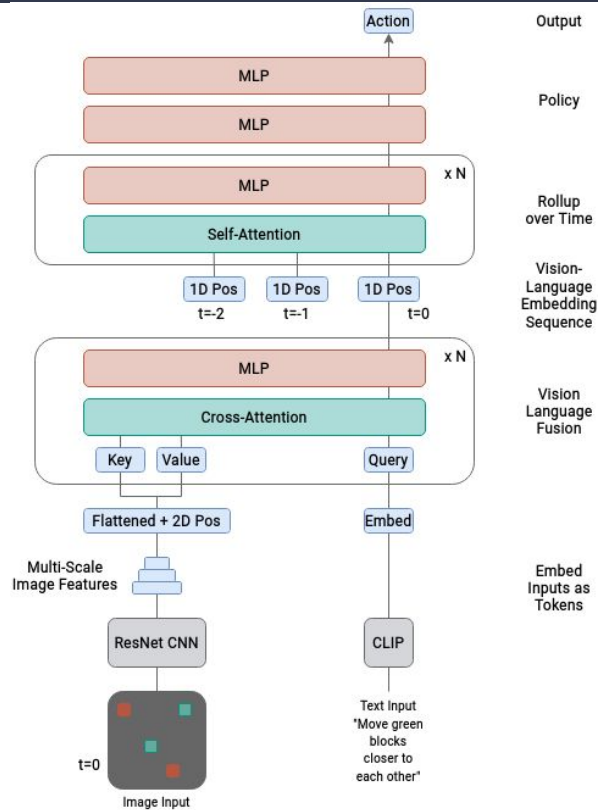
## Image encoding

Pre-trained ResNet + learned convolutions

## Text encoding

Fine-tuned CLIP encoder

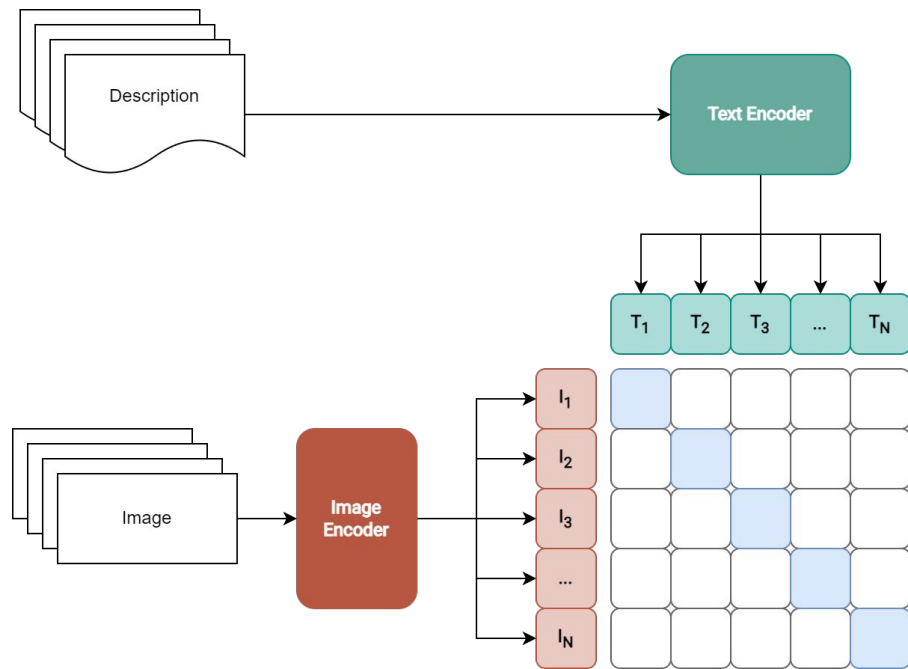
(image, language) → (video, language)



# Relating text with images (CLIP)

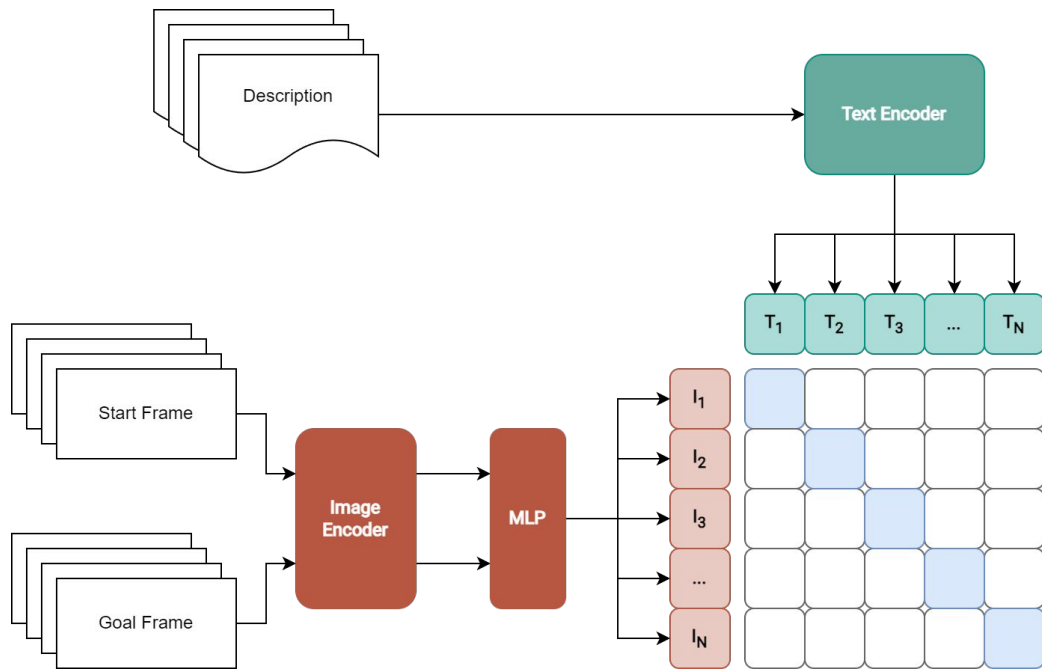
Learning Transferable Visual Models  
From Natural Language Supervision

*Radford, Kim et al (Feb 2021)*



# Relating text with images (CLIP)

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From Natural Language Supervision  
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# Architecture: LAVA (Language Attends to Vision to Act)

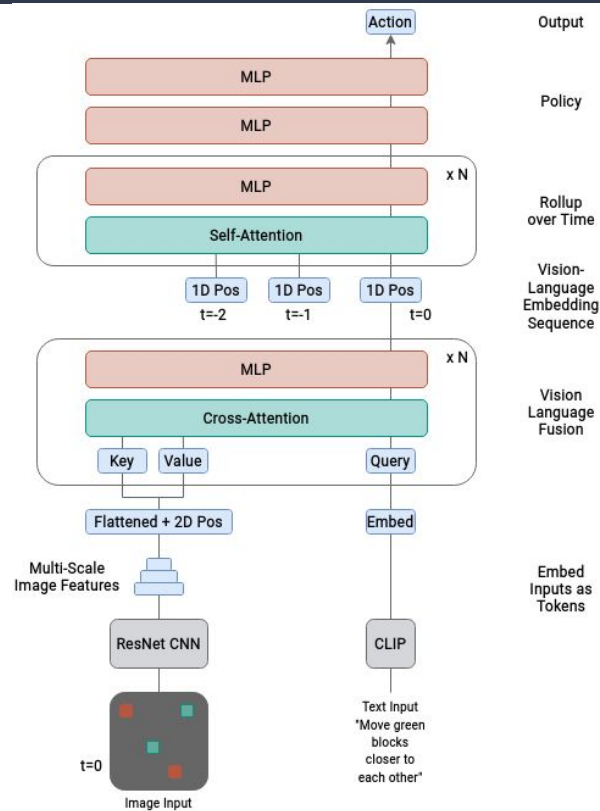
## Decoder-only Transformer

Cross-attention

Multi-Layer Perceptron

## Output

Vision-language embedding for certain timestep



# Architecture: LAVA (Language Attends to Vision to Act)

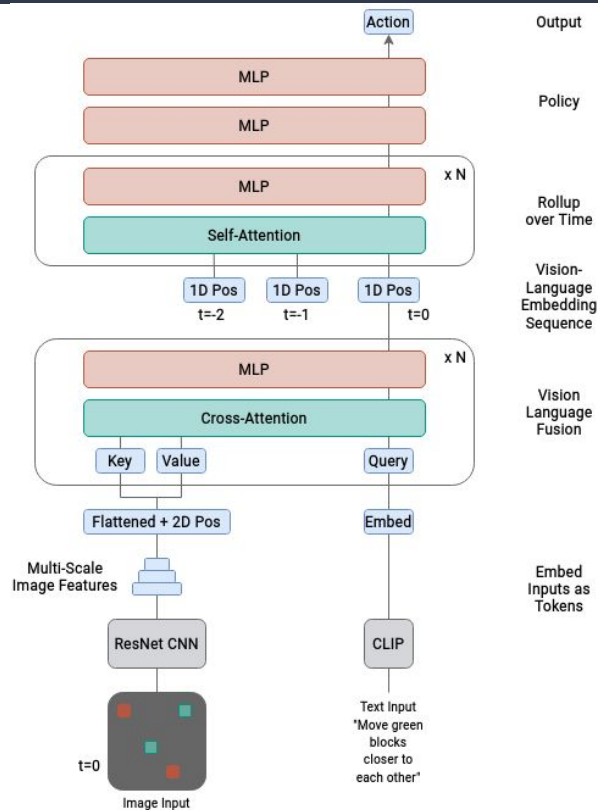
## Encoder-only Transformer

Self-attention

Multi-Layer Perceptron

## Input

Sequence of vision-language embeddings



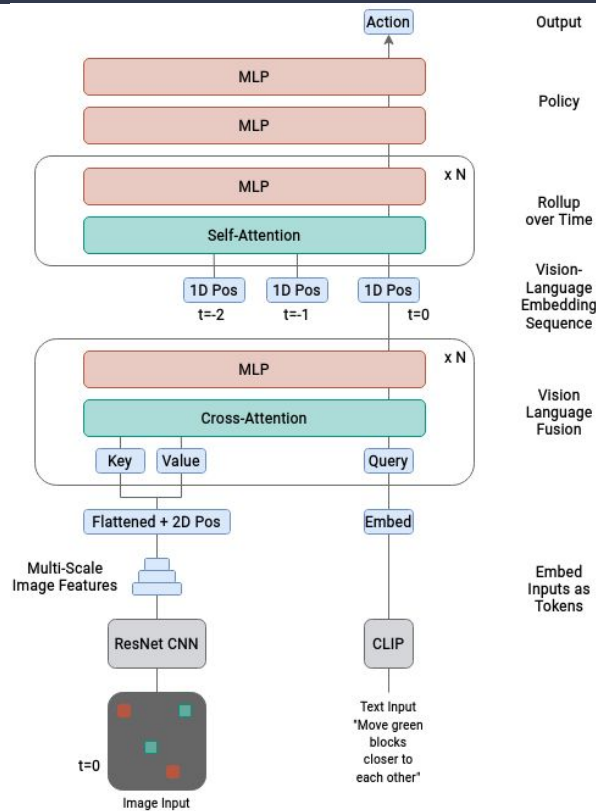
# Architecture: LAVA (Language Attends to Vision to Act)

## Policy

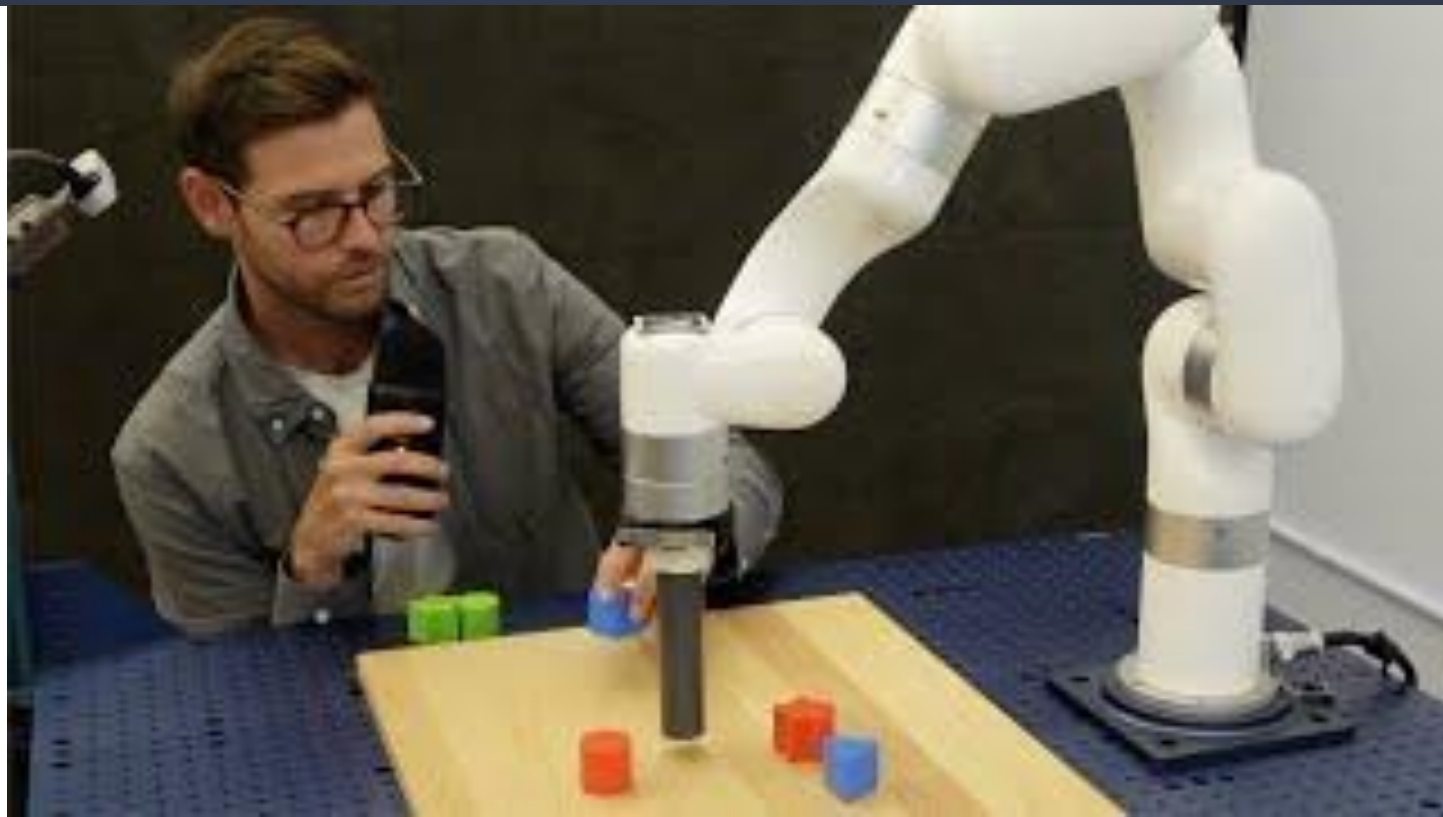
MLP blocks with residual connections

Outputs action

$$\min_{\theta} \sum_{s, a, l \sim \mathcal{D}} \|a, \pi_{\theta}(s, l)\|_2^2$$



# Demonstrations



# Summary

What do we learn from this?

**Language embeddings are useful task descriptors**

**Progress in NLP and CV provides powerful architectures**

**Another use case for attention/transformers**

PaLM-E

# Goal

PaLM-E: An Embodied Multimodal  
Language Model

*Driess, Danny et al. (2023)*

## Visual-language embodied robot

Grounding: connect LLM representations to the real world

Interleave **multi-modal inputs** into **same input embedding space**

### Robot Initiative



Given **<img>**. Q: If a robot wanted to be useful here, what steps should it take?

A: clean table..pick up trash.. pick up chairs.. wipe chairs.. put chairs down.

# Architecture

## Input

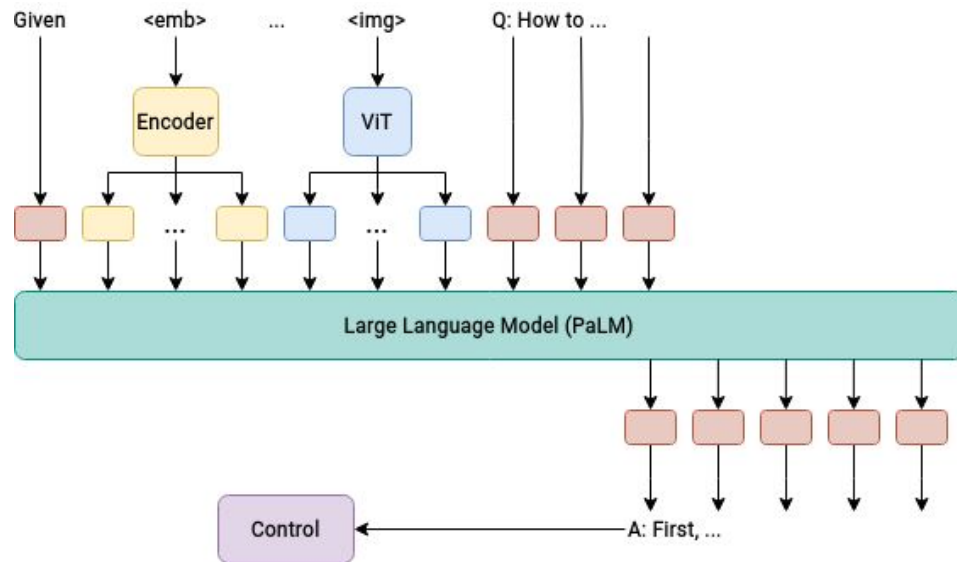
Language, images, sensor readings, ...

Encoders embed all inputs into same space

## Output

Language

Can be used to give short-term instructions to  
language-conditioned policies



# Example Tasks: Visual QA

| Task                          | Given                                                                                                                          | Target                                                 |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Object-table relation</b>  | <img>. Q: Is the red object left, right, or center of the table?                                                               | A: The red object is in the center of the table.       |
| <b>Object-object relation</b> | <img>. Q: Is the yellow object below the blue object?                                                                          | A: No, the yellow object is not below the blue object. |
| <b>Plan feasibility</b>       | <img>. Q: Is it possible to first grasp the blue object, then place it on the yellow object, and then grasp the yellow object? | A: No, this is not possible.                           |

# Example Tasks: Planning

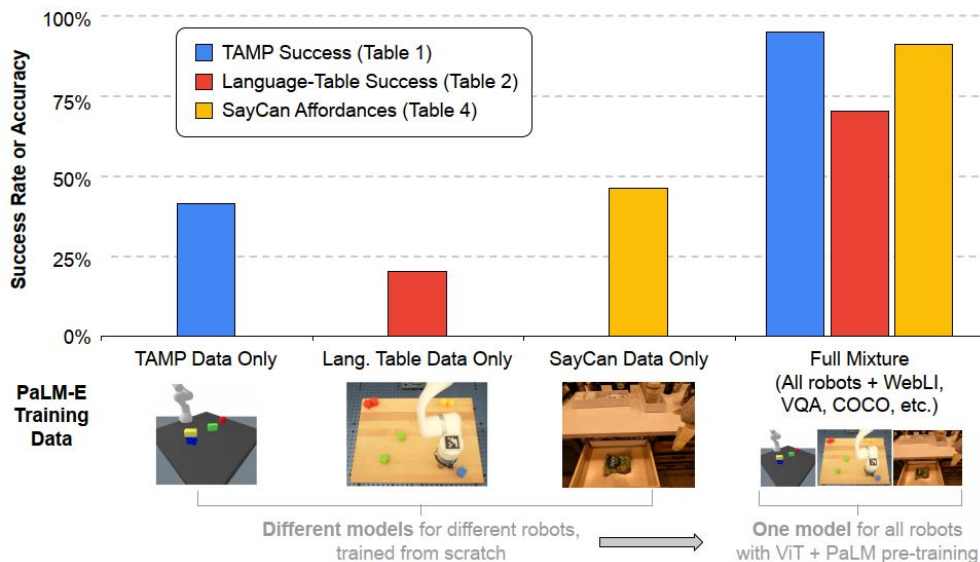
| Task            | Given                                                             | Target                                                                                                                 |
|-----------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Grasping</b> | <img>. Q: How to grasp the green object?                          | A: First grasp the orange object and place it on the table, then grasp the green object.                               |
| <b>Stacking</b> | <img>. Q: How to stack the white object on top of the red object? | A: First grasp the green object and place it on the table, then grasp the white object and place it on the red object. |

# Positive Transfer

PaLM-E: An Embodied Multimodal  
Language Model

*Driess, Danny et al. (2023)*

## Model benefits from multi-task learning



# Positive Transfer

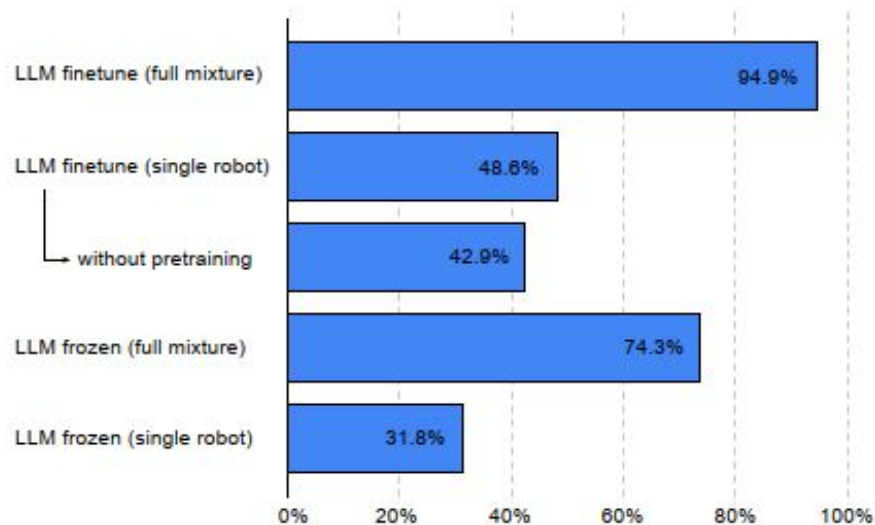
PaLM-E: An Embodied Multimodal  
Language Model

*Driess, Danny et al. (2023)*

Model benefits from multi-task learning

**Training on language tasks helps control**

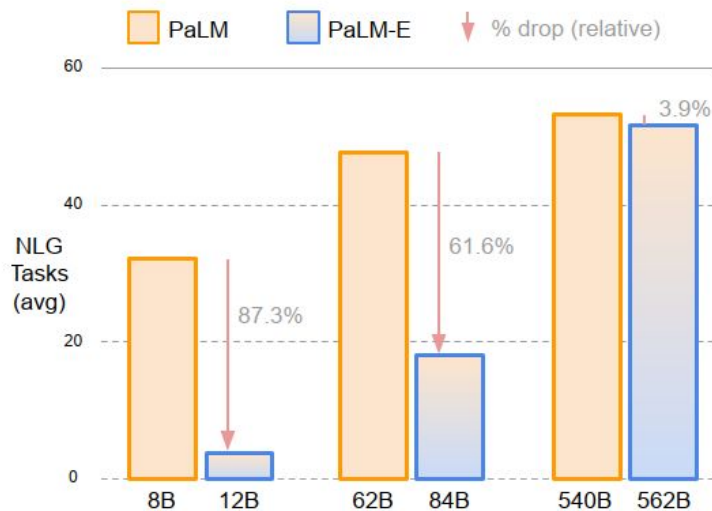
**Only few robotics examples are needed**



# Positive Transfer

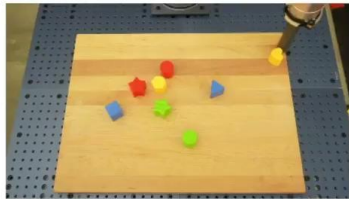
PaLM-E: An Embodied Multimodal Language Model  
*Driess, Danny et al. (2023)*

Model benefits from multi-task learning  
Training on language tasks helps control  
Only few robotics examples are needed  
Large models **preserve NLP capabilities**



# Demos

Given



. Q: How to sort the blocks by colors into the corners? A:

# Demos



Questions?